Making the Graph Neural Network Earn Its Keep

Inductive Prediction of Synthetic Lethality from Gene–Interaction Topology

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One-sentence thesis. Synthetic lethality is a property of the *relationship between two genes' positions in the interaction network* (parallel-pathway redundancy), not of either gene's intrinsic features; a graph neural network therefore has a mechanism to predict it that a feature-only model provably lacks — and this proposal is designed to demonstrate exactly *where* and *why* the graph earns its keep.

1 Motivation

A graph neural network (GNN) is only worth its complexity when the prediction target depends on **graph topology that node features cannot encode**. Many strong student projects train a multilayer perceptron (MLP) or gradient- boosted trees on per-entity feature vectors and obtain excellent accuracy; in those settings a GNN often *matches but does not beat* the simpler model, leaving the graph machinery unjustified. My earlier gut–brain knowledge-graph work ran into precisely this issue.

To avoid that trap, I select a task where the label is *intrinsically relational*. **Synthetic lethality (SL)** is a pair of genes whose simultaneous inhibition kills a cell, while inhibiting either gene alone is tolerated. SL is the central computational strategy in precision oncology for attacking "undruggable" oncogenes (e.g. *KRAS*): rather than drug the oncogene, one drugs a gene that is lethal *only* in the oncogene-mutant background. SL arises from **parallel-pathway redundancy**, a topological motif (Fig. 1), making it an ideal stress test for graph learning.

Methodological novelty. (i) An *inductive* GNN that predicts SL for genes *never seen during training* (cold-gene generalization), where memorization baselines collapse; (ii) a *topology-removal ablation* that falsifies feature leakage by destroying graph structure; and (iii) a *degree-matched negative-sampling* protocol that removes the popularity shortcut which otherwise lets a feature model "cheat."

2 Alignment with the SRI Winning Formula

Every top SRI paper combines six ingredients. Table 1 maps each to a concrete commitment in this project.

3 Problem Formalization

Let $\mathcal{G} = (V, E)$ be an undirected gene–interaction graph with genes V and interaction edges E (physical, regulatory, co-expression). For a cell line, let $\phi(\cdot) \in \{\text{viable}, \text{lethal}\}$ denote the viability of a gene-knockout intervention. A gene pair (i, j) is **synthetic lethal** when

$$\phi(\mathrm{KO}_i) = \text{viable}, \quad \phi(\mathrm{KO}_j) = \text{viable}, \quad \phi(\mathrm{KO}_i \wedge \mathrm{KO}_j) = \text{lethal}. \tag{1}$$

Winning ingredient	Commitment in this project
Real, sizable public data	SynLethDB via KG4SL: ~24k genes, ~476k gene–gene interaction edges, ~73k curated SL pairs.
Clear methodological novelty	Inductive cold-gene SL prediction; topology-removal ablation; degree-matched negatives.
Rigorous evaluation	Baselines incl. **XGBoost**; leakage-safe cold-gene/group splits; 5 seeds; imbalance-aware metrics (AUPRC, Hits@K, MRR).
Interpretability → discovery	GNNExplainer / attention over the SL subgraph; recover known *KRAS* SL partners; rank novel candidates with pathway-redundancy explanations.
Ablations prove each part matters	Topology removal, relation-type ablation, depth, negative-sampling strategy.
Translational / impact story	Candidate SL targets for undruggable oncogenes; a prioritized shortlist for wet-lab follow-up.

Table 1: The proposal is structured around the six elements of strong SRI work.

The learning task is *link prediction on a labeled relation*: given $\mathcal{G}$, predict a score $\hat{s}_{ij} \in [0, 1]$ that pair (i, j) is SL, where the SL relation itself is **excluded** from the message-passing edges E (no label leakage).

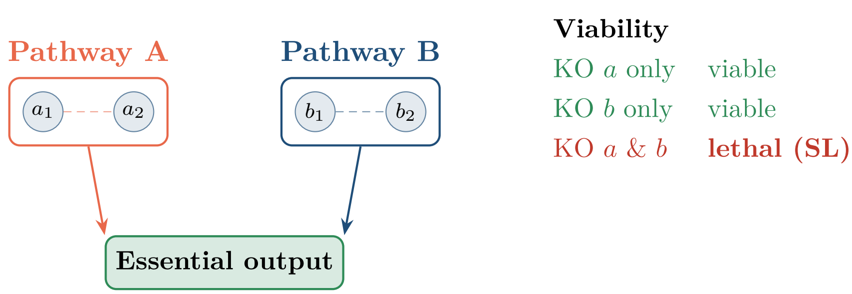

Figure 1: The topological basis of synthetic lethality. Two redundant pathways (A,B) independently sustain an essential output. Removing one is buffered by the other; removing both is lethal. A gene in A and a gene in B are synthetic-lethal — a fact about their *network positions*, invisible to a model that sees only per-gene features.

4 Research Questions & Hypotheses

RQ1

Does encoding the gene-interaction graph let a GNN predict SL better than strong non-graph baselines (XGBoost, MLP, KGE)?

RQ2

Does the GNN advantage persist for *cold genes* absent from training, where memorization-based baselines cannot generalize?

RQ3

Is the GNN advantage *causally* attributable to topology, or to confounds (node features, degree popularity)?

Hypotheses

H1. Transductively, GNN $\approx$ random-walk embeddings $\gg$ MLP/XGBoost.
H2. Inductively (cold-gene), GNN $\gg$ all baselines, which fall to chance.
H3. Destroying topology (rewire/empty graph) collapses the GNN to the feature-only baseline, proving the advantage is structural.

5 Data

Resource	Content	Role
SynLethDB 2.0 (via KG4SL)	~73k human SL gene pairs	Positive labels
KG4SL knowledge graph	~476k gene–gene edges (interaction, regulation, co-expression relations)	Message-passing graph
Explicit non-SL pairs	~3k experimentally non-lethal pairs	Hard negatives (seed)
Degree-matched samples	Negatives drawn to match positive degree distribution	Shortcut-free negatives

Table 2: All data are public and de-identified. The SL/non-SL relations are removed from the message-passing graph to prevent leakage.

Controlled synthetic benchmark. In parallel I will build a generator in which SL is *defined* by topology (genes in different redundant modules of the same essential process), with deliberately uninformative node features. Because the only signal is structural, this isolates the GNN's contribution and powers the falsification ablation.

6 Methods

6.1 Model (ours): inductive heterogeneous GNN

A GraphSAGE encoder produces a gene embedding by aggregating its neighborhood over L=3 message-passing layers:

$$h_v^{(l)} = \sigma\Big(W^{(l)}\big[\,h_v^{(l-1)} \,\big\|\, \mathrm{AGG}_{u\in\mathcal{N}(v)}\, h_u^{(l-1)}\,\big]\Big), \qquad z_v = h_v^{(L)}. \tag{2}$$

Because Eq. (2) reads only the local neighborhood (not a per-node lookup table), embeddings transfer to genes unseen in training — the property that enables cold-gene prediction. A *symmetric* decoder scores a pair (SL is undirected):

$$\hat{s}_{ij} = \mathrm{sigmoid}\Big(\,\mathrm{MLP}\,\big[\,z_i + z_j \,\big\|\, |z_i - z_j|\,\big]\Big). \tag{3}$$

Training minimizes binary cross-entropy over positives $\mathcal{P}$ and sampled negatives $\mathcal{N}$:

$$\mathcal{L} = -\sum_{(i,j)\in\mathcal{P}} \log \hat{s}_{ij} - \sum_{(i,j)\in\mathcal{N}} \log\big(1-\hat{s}_{ij}\big). \tag{4}$$

6.2 Baselines (fair, shared trainer)

6.3 Leakage-safe splits

- **Transductive:** pairs split randomly; all genes seen in training.

Model	Uses graph?	What it tests
GNN (ours)	yes (message passing)	topology-aware inductive prediction
XGBoost	no	strong tabular baseline on pair features
MLP	no (features only)	structure-blind control
node2vec + decoder	structure via memorized walks	transductive-only structure
TransE / DistMult (KGE)	structure via memorized embeddings	shallow KG baselines

Table 3: The MLP and XGBoost share the pair-feature construction; node2vec/KGE share the decoder. Any gap reflects the value of message passing alone.

- **Inductive (cold-gene):** whole *genes* held out; test pairs touch unseen genes, which are removed from the training message-passing graph and only reattached at evaluation. This is the leakage-safe "group split" analogue and the decisive test of generalization.

6.4 Degree-matched negatives (removing the shortcut)

With uniform-random negatives, SL genes (well-studied hubs) have higher degree than random genes, so a single degree feature separates classes — a confound that lets a feature model appear competitive. I instead sample negatives whose endpoint degrees match the positives:

$$\Pr\big[(i',j') \in \mathcal{N}\big] \;\propto\; \nVdash\big[\, b(i') = b(i) \;\wedge\; b(j') = b(j) \,\big]\,,$$

where $b(\cdot)$ is a node's degree-quantile bin and (i,j) is a template positive. This forces reliance on *which* genes interact, not *how connected* they are.

6.5 Metrics (imbalance-aware)

Area under the precision–recall curve (AUPRC) as the primary metric, plus AUROC, and ranking metrics that mirror wet-lab triage of a candidate shortlist:

$$\text{Hits@}K = \frac{1}{|\mathcal{P}|}\sum_{p\in\mathcal{P}} \nVdash[\text{rank}(p) \le K], \qquad \text{MRR} = \frac{1}{|\mathcal{P}|}\sum_{p\in\mathcal{P}} \frac{1}{\text{rank}(p)}. \tag{5}$$

All results are reported as mean $\pm$ std over 5 seeds.

7 Model Pipeline

Figure 2 contrasts the two information pathways under test: the GNN consumes the interaction graph, whereas the feature baselines (XGBoost, MLP) never see an edge. Because every model shares the same decoder and training loop, any performance gap is attributable to message passing alone.

8 Ablations & Interpretability

8.1 Falsification: topology-removal ablation

The headline experiment trains the *identical* GNN on three graphs and compares against the structure-blind MLP:

Additional ablations: per-relation removal (which interaction type matters), message-passing depth (1–4 layers), and negative-sampling strategy (random vs. degree-matched).

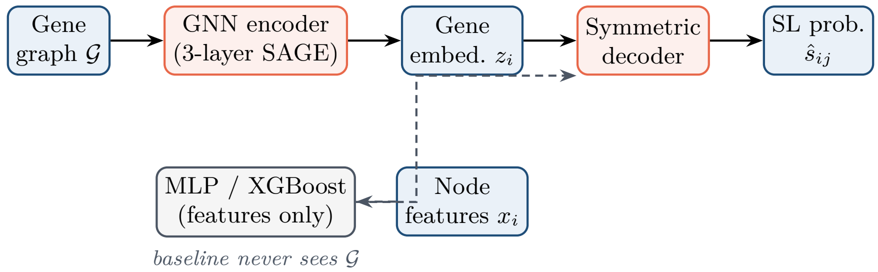

Figure 2: The GNN (orange) propagates over $\mathcal{G}$; the feature baselines (gray) bypass the graph entirely. The comparison isolates the contribution of message passing.

Condition	Graph given to the GNN	Expected
Intact	true interaction graph	high
Rewired	random graph, same node/edge count	$\approx$ chance
Empty	no edges (GNN $\rightarrow$ MLP)	$\approx$ chance

Table 4: If the GNN's advantage is real, destroying structure must erase it.

8.2 Interpretability that yields a discovery

Using GNNExplainer/attention, I will extract the small subgraph most responsible for each high-confidence SL prediction and verify it corresponds to a parallel-pathway motif. As a concrete validation I will check recovery of **known *KRAS* synthetic-lethal partners**, then produce a ranked shortlist of *novel* candidate SL partners, each accompanied by an interpretable pathway-redundancy explanation.

9 Expected Outcomes & Success Criteria

This is a proposal; the table states *targets and qualitative expectations*, not measured results.

Regime / artifact	Expected pattern (success criterion)
Transductive	GNN $\approx$ node2vec $\gg$ XGBoost/MLP: structure carries the signal.
Inductive (cold-gene)	GNN $\gg$ all baselines, which approach chance: only the GNN generalizes structurally.
Topology ablation	Intact $\gg$ rewired $\approx$ empty $\approx$ MLP: the advantage is causal.
Interpretability	Recover known *KRAS* SL partners; deliver a ranked novel shortlist.
Honesty clause	A null result (GNN $\approx$ baselines) will be reported as-is; the task is constructed so a correct GNN *can* win.

Table 5: Success is defined by the *pattern* across regimes, not a single headline number.

10 Eight-Week Timeline

11 Reproducibility, Tooling, and Ethics

Implementation uses **PyTorch Geometric**, the leading GNN library (heterogeneous graphs, link-prediction utilities, neighbor sampling for scale). All code, configs, seeds, and a data manifest

Weeks	Milestones
1–2	Data pipeline (KG4SL download, leakage-free graph, degree-matched negatives); synthetic generator.
3–4	Models in PyG: GNN encoder/decoder; baselines XGBoost, MLP, node2vec, KGE; shared trainer.
5	Transductive + inductive experiments, 5 seeds, imbalance-aware metrics.
6	Ablations: topology removal, relation type, depth, negatives.
7	Interpretability + *KRAS* case study; candidate shortlist.
8	Figures, tables, manuscript; reproducibility package.

Table 6: Schedule for the eight-week SRI cohort.

(accession, version, license, date) will be released. No human subjects are involved; all inputs are public and de-identified. Predictions are research-only and hypothesis-generating, to be confirmed by wet-lab follow-up; no clinical claims are made.

Selected References

1. Hamilton, Ying, Leskovec. *Inductive Representation Learning on Large Graphs (GraphSAGE)*. NeurIPS, 2017.
2. Wang et al. *SynLethDB 2.0: a web-based knowledge graph database on synthetic lethality*. Database, 2022.
3. Zheng et al. *KG4SL: knowledge graph neural network for synthetic lethality prediction in human cancers*. Bioinformatics, 2021.
4. O'Neil, Haber, Buckanovich. *Synthetic lethality and cancer*. Nature Reviews Genetics, 2017.
5. Ying et al. *GNNExplainer: Generating Explanations for Graph Neural Networks*. NeurIPS, 2019.